# Zixian Wang (Purmortal/王梓先)

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## EDUCATION

**Ph.D., Astronomy and Astrophysics** *2019 - 2023*
The University of Sydney (USYD), Sydney, Australia
Thesis: *A Study of Galactic and Extragalactic Evolution with the MUSE Integral-Field Spectrograph*
Supervisor: Joss Bland-Hawthorn, Michael Hayden & Sanjib Sharma

**B.Sc., Astronomy and Astrophysics** *2015 - 2019*
Nanjing University (NJU), Nanjing, China
Thesis: *Comparison of Planetary Properties in Single-Star and Multiple-Star Systems*
Supervisor: Jiwei Xie

## PROFESSIONAL EXPERIENCE

**Parental Leave** *Apr 2026 - Present*

**Postdoctoral Research Associate** *Nov 2023 - Sept 2025*
Department of Physics & Astronomy, University of Utah
Advisor: Anil Seth

**Research Associate (Internship)** *Aug 2023 - Oct 2023*
Astronomy Data and Compute Services (ADACS), Swinburne University of Technology
Supervisor: Jarrod Hurley

## PUBLICATION SUMMARY

**24** papers in total, including **5** papers and **1** proceedings as the first author.
One first-author paper is currently under revision at *Nature Astronomy*.

**Metrics:** h-index = **11** | Total Citations: **497**
**Focus:** Machine learning, Galactic/extragalactic archaeology, globular clusters, interstellar medium.

Full publication list available via SciX.

## OBSERVATIONAL TELESCOPE TIME

**Survey / service observing**
The GALAH survey (AAT/HERMES) – survey observation, $\sim$ 200 hours; $200,000 AUD, 2020–2023

**Principal Investigator**
VLT/MUSE – "The View from Baade's Window: The Star Formation History and Chemo-dynamics of the Inner Galaxy", 9.8 hours (including overhead), $150,000 AUD

**Co-Investigator**
JWST – "Astrometric monitoring of the fast stars around the intermediate-mass black hole in Omega Centauri", 8.6 hours
VLT/FLAMES – "Deciphering Omega Centauri's Formation History by Connecting Stellar Ages and Abundances", 2 nights, $300,000 AUD
ALMA – "The Hunt for Omega Cen*", 13.2 hours
VLT/MUSE – "GECKOS: Turning galaxy evolution on its side with deep observations of edge-on galaxies", 317 hours, $4,800,000 AUD

## SUPERVISION AND TEACHING

### Student Supervision

Callie Clontz (Ph.D., University of Utah), research mentor, 2 papers+1 in prep. *Nov 2023 – May 2025*
Selin (Lily) Erickson (Undergraduate, University of Oklahoma), 1 paper in prep. *Jan 2025 – present*

### Teaching

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|---|---|
| ASTR 6570-001 (Guest Lecturer), Utah | *Spring 2024* |
| BUSS6002: Data Science in Business (tutor), Sydney | *Aug 2022 - Jun 2023* |
| OLET1640: Astronomy: from Big Bang to Darkness (tutor), Sydney | *Feb - Jun 2022* |
| OLET1638: Astronomy: from Stars to Black Holes (tutor), Sydney | *Feb - Jun 2022* |
| OLET1636: Astronomy: from Earth to Exoplanets (tutor), Sydney | *Feb - Jun 2022* |
| OLES1602: The Apocalypse in Science and Culture (tutor), Sydney | *Aug - Dec 2021* |
| 1st year physics (tutorial and lab demonstration), Sydney | *Feb 2021 - Jun 2023* |
| 2nd year physics (lab demonstration), Sydney | *Feb 2020 - Dec 2022* |

## ACADEMIC PRESENTATIONS

### Seminars/Colloquiums

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|---|---|
| Invited, University of New South Wales, Sydney, Australia | *2025* |
| Invited, Max-Planck-Institut für Astronomie (MPIA), Heidelberg, Germany | *2025* |
| Invited, Georg-August-Universität Göttingen, Göttingen, Germany | *2025* |
| Invited, Liverpool John Moores University, Liverpool, UK | *2024* |
| ASA ECR Symposium, University of New South Wales, Australia (virtual) | *2023* |
| PhD Graduation Seminar, Shanghai Astronomical Observatory, Shanghai, China | *2023* |
| PhD Graduation Seminar, University of Chinese Academy of Sciences, Hefei, China | *2023* |
| PhD Graduation Seminar, Nanjing University, Nanjing, China | *2023* |
| PhD Graduation Seminar, Beijing Normal University, Beijing, China | *2023* |
| PhD Graduation Seminar, National Astronomical Observatories, Beijing, China | *2023* |
| PhD Graduation Seminar, Yunnan University, Kunming, China | *2023* |
| PhD Graduation Seminar, Space Telescope Science Institute, Baltimore, USA | *2023* |
| PhD Graduation Seminar, University of Utah, Salt Lake City, USA | *2023* |

### Conference Talks

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| Invited, "GECKOS-Oxford Science Workshop", Oxford, UK | *2024* |
| Contributed, "Wide-Field Spectroscopy vs Galaxy Formation Theory", Tucson, AZ, USA | *2023* |
| Contributed, "DYNAMITE workshop", virtual | *2022* |
| Contributed, "Linking the Galactic and Extra-galactic", Wollongong, Australia | *2022* |
| Contributed, "ACAMAR 8 workshop", virtual | *2022* |
| Contributed, "ACAMAR 7 workshop", virtual | *2021* |
| Recorded, "Spatially Resolved Spectroscopy with Extremely Large Telescopes", virtual | *2021* |
| Contributed, "ASTRO-3D Virtual Science Meeting", virtual | *2021* |
| Recorded, "MAVIS workshop", virtual | *2021* |
| Contributed, "Stromlo Student Seminars", virtual | *2020* |
| Contributed, "1st Australian BlueMUSE Science Workshop", virtual | *2020* |

### Conference Posters

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|---|---|
| "IAU 377, Early Disk-Galaxy Formation From JWST to the Milky Way", virtual | *2023* |
| "2022 XXXIst General Assembly International Astronomical Union", Busan, Korea | *2022* |
| "ASA Annual Science Meeting", Hobart, Australia | *2022* |
| "ASA Annual Science Meeting", virtual | *2021* |
| "European Astronomical Society Annual Meeting", virtual | *2021* |
| "The University of Sydney HDR symposium", Sydney, Australia | *2020* |

## SOFTWARE

**GalCraft**
Sole Developer. A Python module to generate mock integral-field spectrograph observation of the Milky Way using stellar catalogs. See details on the project website.

**mapviewer-web**
Sole Developer. A web version of Mapviewer to illustrate galaxy properties obtained from the GIST pipeline. It is developed using DashPlotly. See details on the project website.

**nGIST**
Contributor. An updated version of GIST for the gist-geckos Pipeline to analyze MUSE data from the GECKOS survey. See details on the project website.

**MCFOST**
Contributor. A 3D continuum and line radiative transfer code based on a hybrid Monte Carlo and ray-tracing method. See details on the project website.

## COLLABORATION

**The GALAH Survey (galah-survey.org)** *2019 - present*
Contributed to AAT/HERMES survey observations and research on Galactic archaeology.

**The GECKOS Survey (geckos-survey.org)** *2022 - present*
Lead research on connecting Galactic and extragalactic archaeology, and contributed to the development of extragalactic data-analysis tools.

**oMEGACat (omegacatalog.github.io)** *2023 - present*
Lead research on the chemical evolution history and interstellar/intracluster medium of $\omega$ Centauri, and contributed to studies of multiple stellar populations and helium enrichment, etc.

## MEDIA AND OUTREACH

Sydney observatory tour for Chinese new year @ Powerhouse Museum *2023*
Measuring reliable stellar abundances towards crowded regions using MUSE @ AAL *2022*
Explore the Double Asteroid Redirection Test (DART) mission (in Mandarin) @ SBS Podcast *2022*

## ACADEMIC SERVICES

Evaluator, "Undergraduate Research Summer Symposium", UUtah *Summer 2024*
LoC, "Linking the Galactic and Extragalactic", Wollongong *Nov - Dec 2022*
Organizer, Milky Way / Extragalactic Journal Club, USYD *Jun 2021 - Feb 2023*
Seminar & Visitor host, SIfA seminar committee, USYD *Jan 2020 - Jun 2023*

## AWARDS AND HONORS

ASA Travel Grant, $1,400 AUD *Sydney, 2022*
USYD HDR Completion Scholarship, $7,500 AUD+tuition fee *Sydney, 2022*
ASTRO3D Travel Grant, $9,000 AUD *Sydney, 2020-2022*
Postgraduate Research Support Scheme (PRSS), $5,000 AUD *Sydney, 2020-2022*
Second-Year Physics Laboratory Teaching Award *Sydney, 2021*
ASTRO-3D Postgraduate Scholarship *Sydney, 2019*
USYD Postgraduate Scholarship *Sydney, 2019*
Chenxue Scholarship for Overseas Study *Nanjing, 2019*
Outstanding Undergraduate Student Award *Nanjing, 2019*
Top-norch Program Scholarship *Nanjing, 2019*
National Encouragement Scholarship *Nanjing, 2018*
Top-norch Program Scholarship *Nanjing, 2015*

## FULL PUBLICATION LIST

### Refereed Journal Articles — Published or Accepted

*First-author & Major Contributions*

1. **Zixian Wang**, Anil C. Seth, Callie Clontz, Nadine Neumayer, Maximilian Häberle, et al., "oMEGA-Cat. IX. Chemical Tagging of Omega Centauri Populations with Machine-Learning-Inferred Abundances from the MUSE Spectrograph", 2026, arXiv:2603.01041 *(accepted for publication in ApJ)* [SciX]

2. **Zixian Wang**, Anil C. Seth, Marilyn Latour, Jay Strader, Maximilian Häberle, et al., "oMEGA-Cat. VII. Tracing Interstellar and Intracluster Medium of $\omega$ Centauri using Sodium Absorptions", 2025, ApJ, 994, 143 [SciX]

3. **Zixian Wang**, Sanjib Sharma, Michael R. Hayden, Jesse van de Sande, Joss Bland-Hawthorn, Sam Vaughan, Marie Martig, et al., "Validating full-spectrum fitting with a synthetic integral-field spectroscopic observation of the Milky Way", 2024, MNRAS, 534, 1175 [SciX]

4. **Zixian Wang**, Michael R. Hayden, Sanjib Sharma, Maosheng Xiang, Yuan-Sen Ting, Joss Bland-Hawthorn, Boquan Chen, "Reliable stellar abundances of individual stars with the MUSE integral-field spectrograph", 2022, MNRAS, 514, 1034 [SciX]

5. Callie Clontz, Anil C. Seth, **Zixian Wang**, Maximilian Häberle, Maria Selina Nitschai, et al., "oMEGACat. VIII. A Subpopulation Census of $\omega$ Centauri", 2026, ApJ, 998, 177 [SciX]

6. Callie Clontz, Anil C. Seth, **Zixian Wang**, Stefano O. Souza, Maximilian Häberle, et al., "oMEGA-Cat. V. Helium Enrichment in $\omega$ Centauri as a Function of Metallicity", 2025, ApJ, 984, 162 [SciX]

7. Randa Asa'd, Svea Hernandez, J. M. John, Mayte Alfaro-Cuello, **Zixian Wang**, A. As'ad, A. Vasini, Francesca Matteucci, "NGC 1856: Using Machine Learning Techniques to Uncover Detailed Stellar Abundances from MUSE Data", 2024, AJ, 167, 265 [SciX]

*Other Publications*

1. Ding Y., Martig M., Zhu L., Fraser-McKelvie A., Leaman R., et al. (**including Wang Z.**), "The GECKOS survey: assembly history of the lenticular galaxy NGC 3957", 2026, MNRAS, 552, stag1634 [SciX]

2. Stefano Souza, Nadine Neumayer, Anil C. Seth, **Zixian Wang**, Callie Clontz, et al., "oMEGACat. X. Shedding Light on the Disrupted Dwarf Galaxy of Omega Centauri", 2026, ApJ, 1008, 158 [SciX]

3. Maximilian Häberle, Dmitry Bizyaev, Alina Boecker, Callie Clontz, Bruno Dias, et al. (**including Zixian Wang**), "SDSS-V Local Volume Mapper (LVM): The integrated light and internal rotation of Omega Centauri", 2026, A&A, 713, A301 [SciX]

4. Matthew Whitaker, Evan Kerr, Anil C. Seth, Maximilian Häberle, Jay Strader, et al. (**including Zixian Wang**), "A Long Period Stellar-mass Black Hole Binary in $\omega$ Centauri", 2026, ApJ, 1006, L1 [SciX]

5. Benjamin J. Gibson, Gail Zasowski, Anil C. Seth, Dimitri A. Gadotti, **Zixian Wang**, et al., "Towards understanding the Milky Way's typicality: assessing the chemodynamics of M31's bulge and bar, thick and thin discs", 2025, MNRAS, 542, 669 [SciX]

6. Amelia Fraser-McKelvie, Jesse van de Sande, Dimitri A. Gadotti, Eric Emsellem, Toby Brown, et al. (**including Zixian Wang**), "The GECKOS survey: Identifying kinematic sub-structures in edge-on galaxies", 2025, A&A, 700, A237 [SciX]

7. Sven Buder, Janez Kos, Ella X. Wang, Madeleine McKenzie, Madeline Howell, et al. (**including Zixian Purmortal Wang**), "The GALAH survey: Data release 4", 2025, PASA, 42, e051 [SciX]

8. Maximilian Häberle, Nadine Neumayer, Callie Clontz, Anil C. Seth, Peter J. Smith, et al. (**including Zixian Wang**), "oMEGACat. VI. Analysis of the Overall Kinematics of Omega Centauri in 3D: Velocity Dispersion, Kinematic Distance, Anisotropy, and Energy Equipartition", 2025, ApJ, 983, 95 [SciX]

9. Callie Clontz, Anil C. Seth, Aaron Dotter, Maximilian Häberle, Maria Selina Nitschai, Nadine Neumayer, Anja Feldmeier-Krause, et al. (**including Zixian Wang**), "oMEGACat. IV. Constraining the Ages of Omega Centauri Subgiant Branch Stars with HST and MUSE", 2024, ApJ, 977, 14 [SciX]

10. Maria Selina Nitschai, Nadine Neumayer, Maximilian Häberle, Callie Clontz, Anil C. Seth, et al. (**including Zixian Wang**), "oMEGACat. III. Multiband Photometry and Metallicities Reveal Spatially Well-mixed Populations within $\omega$ Centauri's Half-light Radius", 2024, ApJ, 970, 152 [SciX]

11. Michael R. Hayden, Sanjib Sharma, Joss Bland-Hawthorn, Lorenzo Spina, Sven Buder, et al. (**including Purmortal Wang**), "The GALAH survey: chemical clocks", 2022, MNRAS, 517, 5325 [SciX]

12. Sanjib Sharma, Michael R. Hayden, Joss Bland-Hawthorn, Dennis Stello, Sven Buder, et al. (**including Purmortal Wang**), "The GALAH Survey: dependence of elemental abundances on age and metallicity for stars in the Galactic disc", 2022, MNRAS, 510, 734 [SciX]

**Articles Submitted to Refereed Journals**

1. Sadaf Z. Aliakbarzadeh, Caroline Foster, Tayyaba Zafar, Sarah L. Martell, Sebastian Kamann, et al. (**including Zixian Wang**), "The GECKOS Survey: Identifying Globular Clusters and H II Regions as Kinematic Tracers in Disc Galaxies Using Spatially Resolved Spectroscopy", 2026, submitted to MNRAS.

2. Peter J. Smith, Nadine Neumayer, Anil C. Seth, Callie Clontz, Maximilian Häberle, et al. (**including Zixian Wang**), "oMEGACat. XI. Kinematic Characterization of Omega Centauri's Subpopulations: A Flattened Primordial Component", 2026, submitted to ApJ.

3. Maximilian Häberle, Callie Clontz, Massimo Griggio, Mattia Libralato, Isabella Carlton et al. (**including Zixian Wang**), "oMEGACat-JWST I.: Project Description and Photometric Data Release", 2026, submitted to A&A.

4. **Zixian Wang**, Michael R. Hayden, Sanjib Sharma, Joss Bland-Hawthorn, Anil C. Seth, Gail Zasowski, "Age, Chemistry, and Kinematics of the Inner Galaxy Revealed by MUSE", 2025, arXiv:2512.09527 *(submitted to Nature Astronomy; under revision)* [SciX]

**Conference Proceedings**

1. **Zixian Wang**, Michael Hayden, Sanjib Sharma, Joss Bland-Hawthorn, Jesse van de Sande, Sam Vaughan, "The Milky Way in context: Building an IFS datacube of the Galaxy", 2024, Proceedings of the International Astronomical Union, 20, 265–269 [SciX]

**Software Publications**

1. Amelia Fraser-McKelvie, Jesse van de Sande, et al., (**including Zixian Wang**). "nGIST: The new Galaxy Integral-field Spectroscopy Tool". Astrophysics Source Code Library, 2025, [SciX].

2. **Zixian Wang**, "GalCraft: Building integral-field spectrograph data cubes of the Milky Way". Astrophysics Source Code Library, 2024, [SciX].

## REFERENCES

**Prof. Anil Seth (Postdoc Advisor)**
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